
CSE (DATA SCIENCE)

Major Project : Phase-I
Year & Sem : IV & I Sem

A.Y : 2026-27
Date : 07.07.2026

Title of the project: UPI Guardian: AI-Based Adaptive Fraud Prevention Framework

Project Batch No:04

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Abstract:

The rapid growth of the Unified Payments Interface (UPI) has made digital payments faster, simpler, and more convenient. However, it has also increased frauds such as phishing attacks, fake payment requests, QR code scams, and social engineering. Existing AI-based fraud detection systems can identify suspicious transactions but cannot always prevent users from unknowingly sending money to fraudsters. To address this problem, we propose **UPI Guardian**, an **AI-Based Adaptive Fraud Prevention Framework** that performs AI Risk Analysis using transaction behavior, payment history, receiver trust score, and suspicious SMS indicators. Based on the calculated risk, the system activates **Smart Payment Intent Verification**, **Delayed Protection Mode**, and **AI Explainable Risk Analysis** for high-risk transactions. The framework also provides **Personalized Multilingual Voice Assistance** in English, Hindi, and Telugu to improve accessibility. Overall, UPI Guardian aims to make digital payments safer, smarter, more secure, and user-friendly.

Keywords: UPI, Fraud Prevention, AI Risk Analysis, Trust Score, Smart Payment Intent Verification, Delayed Protection Mode, Explainable AI, Multilingual Voice Assistance.

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